

Virtual Memory

1 Core Idea

Definition

Virtual memory is a memory abstraction that lets each process have its own address space, divided into fixed-size chunks called **pages**.

Not every page must be in physical memory at the same time. The operating system keeps needed pages in RAM and stores the rest on disk.

Big Idea

program can run even if only part of it is in RAM

2 Why Virtual Memory Is Needed

Base and limit registers give each process an address space, but they do not solve the problem of programs and process collections being larger than RAM.

Problem	Explanation
Large programs	A program may need more memory than physical RAM can hold.
Many processes	Each process may fit alone, but all processes together may exceed RAM.
Slow swapping	Swapping whole multi-GB processes to disk can take seconds.

Why Swapping Is Not Enough

Swapping moves whole processes. Virtual memory moves smaller pieces, so the system avoids copying an entire large process every time.

3 Before Virtual Memory: Overlays

Overlay

An **overlay** is a manually separated piece of a large program that is loaded only when needed.

With overlays, the program starts by loading an overlay manager. The manager loads overlay 0, then later loads overlay 1, overlay 2, and so on.

Step	What happens
1	Load the overlay manager.
2	Load and run the first required overlay.
3	Replace or add overlays as execution continues.
4	Keep unused overlays on disk.

Overlay Problem

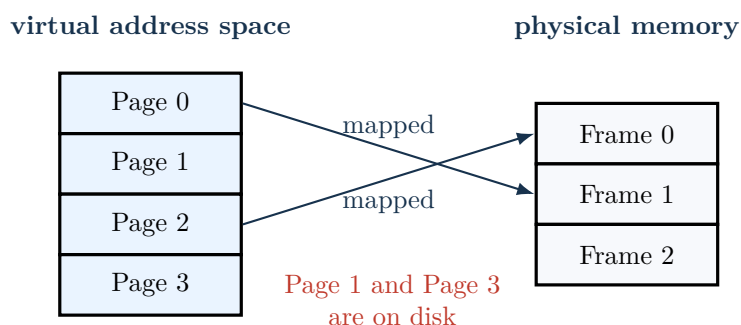
The programmer had to split the program manually. This was time consuming, difficult, and error prone.

4 Pages

Page

A **page** is a contiguous range of addresses inside a process's virtual address space.

Virtual memory splits each process address space into pages. Physical memory is divided into matching slots called page frames.



Key Point

A page can be in RAM or on disk. The process still uses the same virtual addresses either way.

5 How Virtual Memory Runs a Program

When a process references memory, the hardware checks whether the referenced page is already in RAM.

Case	Action
Page is in RAM	Hardware translates the virtual address to a physical address and continues.
Page is not in RAM	A page fault occurs; the operating system loads the missing page from disk.

6 Page Fault

Page Fault

A **page fault** occurs when a process references a virtual page that is not currently in physical memory.

Not every page fault means the same thing. The operating system must check whether the referenced virtual address is valid for the process.

Situation	OS response
Valid page, not in RAM	Load the page from disk into RAM, then restart the instruction.
Invalid address	Raise an error such as a segmentation fault or access violation.

The failed instruction is not simply skipped. After the missing page is loaded, the instruction is normally restarted.



Result

The program behaves as if the page had been in memory all along, except that the page fault causes a delay.

7 Virtual Memory vs Swapping

Method	Unit moved	Main behavior
Swapping	Whole process	A complete process is moved between RAM and disk.
Virtual memory	Page	Only needed pieces of a process are moved.

Main Advantage

Virtual memory is more flexible because RAM can contain pieces of many different processes at the same time.

8 Relation to Base and Limit

Virtual memory generalizes the base-and-limit idea.

Base and limit	Virtual memory
One simple relocation region per process.	Many small pages can be mapped independently.
Whole process usually needs a contiguous memory region.	Pages can be placed in different page frames.
All needed memory should be in RAM.	Only currently needed pages must be in RAM.

9 Multiprogramming Benefit

Virtual memory works well with multiprogramming. RAM can hold pages from many processes at once.

- Process P_1 may have some pages in RAM and others on disk.
- Process P_2 may also have only its active pages in RAM.
- If P_1 waits for a page to be loaded, the CPU can run P_2 .

CPU Utilization

While one process waits for disk I/O, another process can use the CPU.

10 Summary

Concept	Meaning
Virtual memory	Address-space abstraction where only some pages need to be in RAM.
Page	Fixed-size chunk of a virtual address space.
Page frame	Physical-memory slot that can hold one page.
Page fault	Trap caused by referencing a page not currently in RAM; invalid addresses cause an access error.
Main benefit	Programs larger than RAM can run, and many processes can share RAM efficiently.